RAPPORT
Single-Ramp Analog-to-Digital Converter (C1)

1 Exercise Objectives

This investigation addresses Exercise N°1 from the Fundamental Electronics II TD Series. The primary objectives include:

- Determining the digital code corresponding to an analog input voltage $V_A = 0.12\text{V}$
- Analysing voltage waveforms at critical circuit nodes (points A, B, C, D, and E)
- Validating theoretical predictions through circuit simulation

The converter operates with the following parameters:

- **Resolution:** 8 bits (256 discrete levels)
- **Reference Voltage (V_{ref}):** 8V
- **Comparator Saturation Voltage (V_{sat}):** 5V
- **Initial Counter State:** Reset to zero (00000000)
- **Target Sample:** $V_A = 0.12\text{V}$

2 Circuit Description

The single-ramp ADC implements a successive comparison technique, where a digitally controlled staircase voltage is generated and compared against the input analogue sample. The circuit comprises five primary functional blocks:

- **Counter:** Binary counter generating sequential digital codes
- **Digital-to-Analog Converter (DAC):** Converting digital codes to analog reference voltages
- **Comparator:** Determining voltage magnitude relationships
- **Clock Generator:** Providing timing signals
- **Control Logic:** Managing conversion sequencing

3 Component Specifications

3.1 8-Bit Digital-to-Analog Converter DAC0808

The DAC0808 is an 8-bit current-output DAC, configured with an operational amplifier to produce a voltage output. It translates the 8-bit binary value supplied by the counter into a corresponding analog voltage V_E based on a weighted conversion principle.

Key Specifications:

- **Resolution:** 8 bits (256 discrete levels)
- **Reference Voltage Range:** ± 10 V
- **Output Voltage Range:** 0 to V_{ref} (depending on configuration)
- **Supply Voltage:** ± 4.5 V to ± 18 V
- **Reference Current:** 2 mA

In this setup, $V_{ref} = 8$ V, and the DAC input lines **B1 (MSB)** through **B8 (LSB)** are driven by the counter outputs **Q8 to Q1**, with **Q8** serving as the most significant bit.

3.2 Voltage Comparator LM311

Role in the Circuit: The voltage comparator evaluates the analog input voltage V_A at the non-inverting terminal (V+) against the DAC-generated voltage V_E at the inverting terminal (V). Based on this comparison, it produces a binary output V_B, which is used to control and guide the conversion process.

Key Specifications:

- **Supply Voltage:** Operates within a range of 5 V to 36 V
- **Input Offset Voltage:** Typically 2 mV
- **Input Bias Current:** Approximately 100 nA
- **Input Offset Current:** Around 20 nA
- **Common-Mode Rejection Ratio (CMRR):** 100 dB
- **Response Time:** 200 ns
- **Output Voltage Levels:**
  - High Level (V_{sat}): 5 V via pull-up resistor
  - Low Level: 0 V with the open-collector output pulled to ground
- **Strobe Current:** Up to 150 mA (when used)

3.3 8-Bit Binary Counter – CD4020

Component Overview: The CD4020 is a CMOS 14-stage binary counter, configured here to operate as an 8-bit counter.

Function in the Circuit: It produces a continuous sequence of binary values ranging from 00000000 to 11111111. These outputs are fed directly to the DAC inputs, defining the step-by-step timing and progression of the conversion process.

Key Electrical Specifications:

- **Operating Supply Voltage:** 3 V to 18 V
- **Logic Levels ($V_{DD} = 5$ V):**
  - Logic High (1): $V_{IH} \geq 3.5$ V, $V_{OH} \geq 4.95$ V
  - Logic Low (0): $V_{IL} \leq 1.5$ V, $V_{OL} \leq 0.05$ V
- **Maximum Clock Frequency:** 2.5 MHz at 5 V
- **Typical Propagation Delay:** 160 ns
- **Output Current Capability:** 1.5 mA
- **Quiescent Power Consumption:** Approximately 10 µW at 5 V

Operational Description:

The counter advances its binary count on every rising edge of the clock signal applied to the CLK input. Activating the R (reset) input asynchronously forces all outputs to a logic-low state. In this configuration, outputs Q1 through Q8 deliver the 8-bit binary sequence required to drive the DAC.

Timing Parameters:

- **Minimum Setup Time:** 40 ns
- **Minimum Hold Time:** 20 ns
- **Minimum Reset Pulse Width:** 100 ns

3.4 Logic Gate 74LS08

Component: 74LS08 (Quad 2-Input AND Gate, LS-TTL)

Function in Circuit: Controls the clock signal sent to the counter. Clock pulses are passed only when the comparator output V_B is HIGH, stopping the conversion process when $V_A < V_E$.

Specifications:

- **Supply Voltage:** 5 V (±0.25 V)
- **Logic Levels:**
  - Input HIGH (V_{IH}): ≥ 2 V
  - Input LOW (V_{IL}): ≤ 0.8 V
  - Output HIGH (V_{OH}): ≥ 2.7 V

- Output LOW (V_{OL}): ≤ 0.5 V

- **Propagation Delay:** $\sim$15 ns
- **Power Consumption:** $\sim$4.4 mW per gate
- **Fan-out:** Up to 20 LS-TTL loads

Operational Principle: The AND gate combines the system clock with the comparator output V_B. The counter advances only when $V_B = 1$, enabling conditional counting required for the single-ramp ADC operation.

4 The Conversion Process

Step	**Counter**	**Binary Code**	V_E **(V)**	$V_A - V_E$ **(V)**	**Comparator State** (V_B)
0	0	00000000	0.0000	0.12000	5 V
1	1	00000001	0.03125	0.08875	5 V
2	2	00000010	0.06250	0.05750	5 V
3	3	00000011	0.09375	0.02625	5 V
4	4	00000100	0.12500	−0.00500	0 V

Table 1: Conversion process steps showing counter progression and voltage comparisons

5 Simulation

Simulation Tool: Proteus Design Suite Professional (v8.15)

Type of Simulation: Transient analysis incorporating interactive components

5.1 Simulation Waveforms

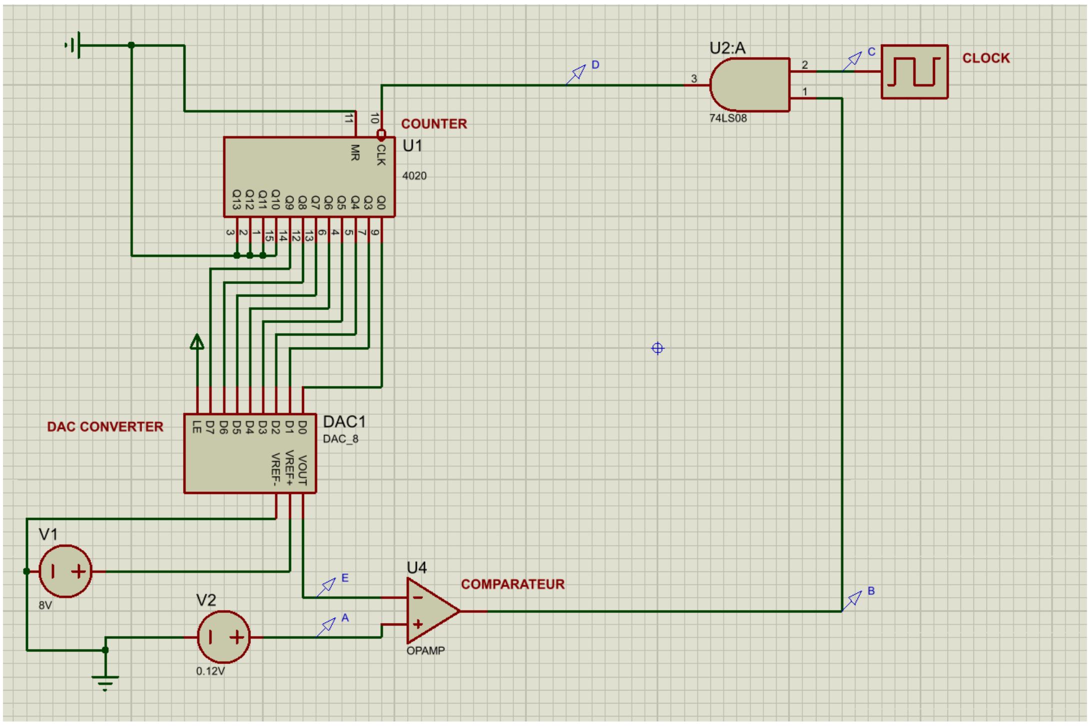
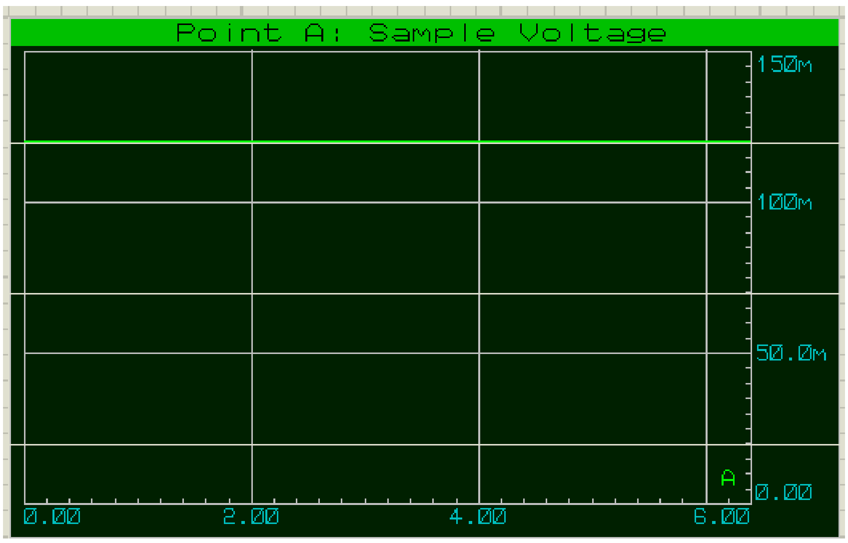


Figure 1: Composite timing diagram showing voltage variations at all key nodes (A, B, C, D, and E)

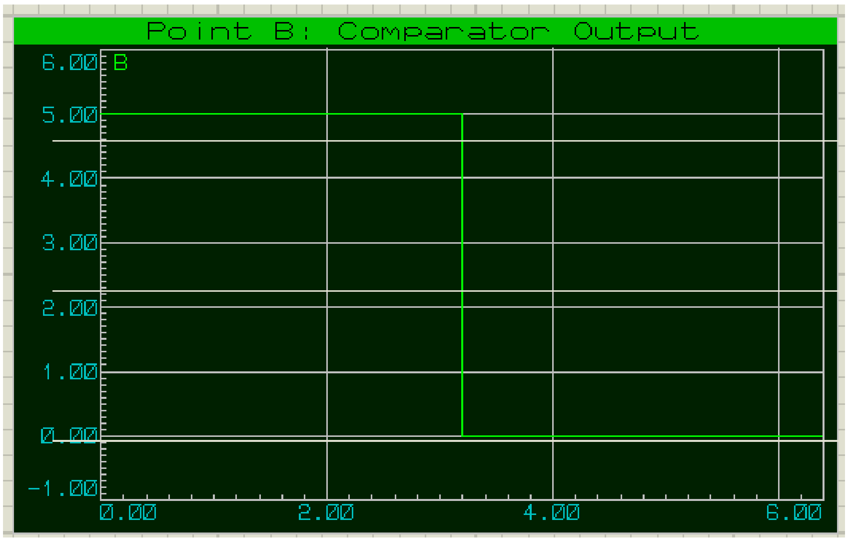
Figure 1 shows a composite timing diagram generated using Falstad, depicting the voltage variations at all key nodes (A, B, C, D, and E) throughout the conversion process. A simplified clock waveform was deliberately employed in this simulation to offer a clear and comprehensive visualization of signal interactions and timing relationships during operation.

Signal Descriptions:

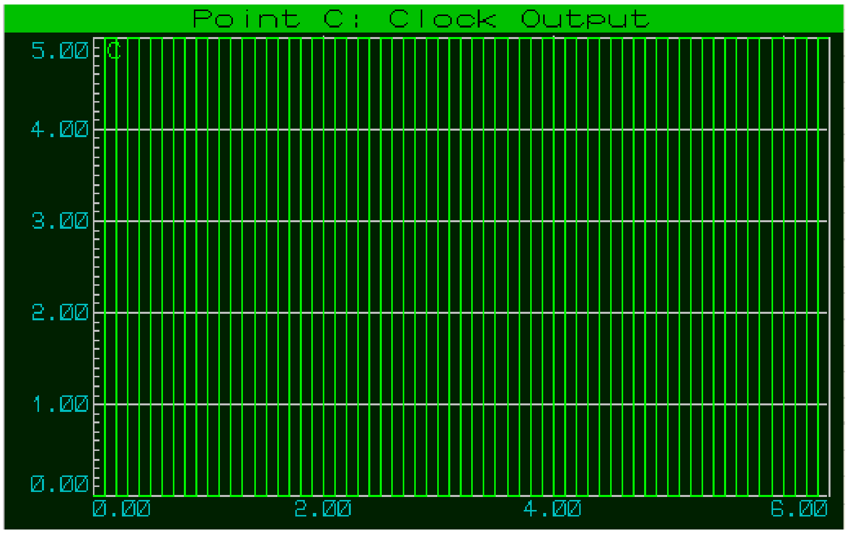
- **A:** Maintains a constant analog input voltage



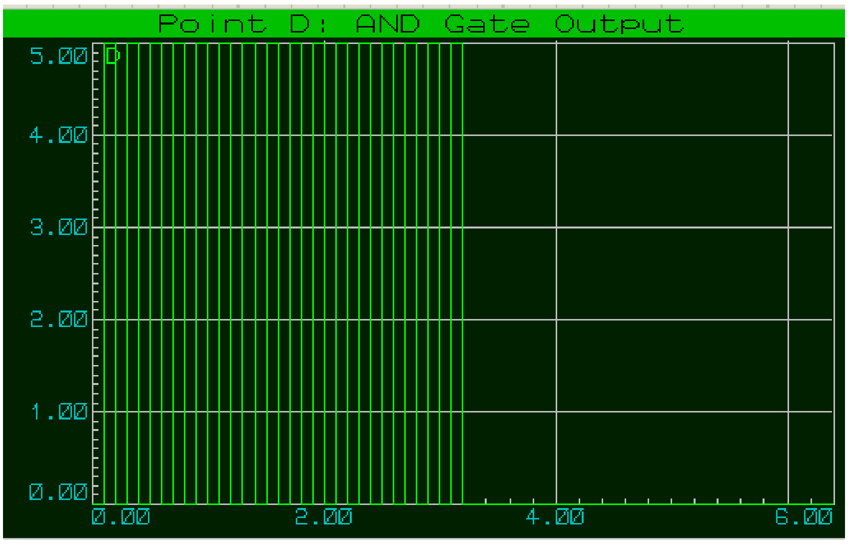
- **B:** Output of the comparator, responsible for controlling the conversion process



- **C:** Clock signal passed through gating logic



- **D:** Control signal applied to the AND gate



- **E:** Staircase output voltage from the Digital-to-Analog Converter (DAC)